

1 Supplementary 1: Equations of the M1 model

The equations are similar to those of the Pay model 2022b. Equations 19 is added along with modifications in Equations 6 and 10 of P51 protein and PIF mRNA, respectively.

$$\frac{d[\text{CLm}]}{dt} = (v_1 + L_a) \left(\frac{1}{1 + \left(\frac{[\text{P97p}]}{K_1} \right)^2 + \left(\frac{[\text{P51p}]}{K_2} \right)^2} \right) - (k_{1L} \Theta_{\text{PhyA}} + k_{1D} (1 - \Theta_{\text{PhyA}})) [\text{CLm}] \quad (1)$$

$$\frac{d[\text{CLp}]}{dt} = (p_1 + p_{1L} \Theta_{\text{PhyA}}) [\text{CLm}] - d_1 [\text{CLp}] \quad (2)$$

$$\frac{d[\text{P97m}]}{dt} = (v_2 + L_b) \left(\frac{1}{1 + \left(\frac{[\text{CLp}]}{K_3} \right)^2 + \left(\frac{[\text{P51p}]}{K_4} \right)^2 + \left(\frac{[\text{ELp}]}{K_5} \right)^2} \right) - k_2 [\text{P97m}] \quad (3)$$

$$\frac{d[\text{P97p}]}{dt} = p_2 [\text{P97m}] - d_{2D} (1 - \Theta_{\text{PhyA}}) [\text{P97p}] - d_{2L} \Theta_{\text{PhyA}} [\text{P97p}] \quad (4)$$

$$\frac{d[\text{P51m}]}{dt} = \left(\frac{v_3}{1 + \left(\frac{[\text{CLp}]}{K_6} \right)^2 + \left(\frac{[\text{P51p}]}{K_7} \right)^2} \right) - k_3 [\text{P51m}] \quad (5)$$

$$\frac{d[\text{P51p}]}{dt} = p_3 [\text{P51m}] - d_{3D} (1 - \Theta_{\text{PhyA}}) [\text{P51p}] - d_{3L} \Theta_{\text{PhyA}} [\text{P51p}] - \frac{d_{p6} [\text{GZp}] [\text{P51p}]}{G_{kp} + [\text{P51p}]} \quad (6)$$

$$\frac{d[\text{ELm}]}{dt} = \left(\frac{v_4 \Theta_{\text{PhyA}}}{1 + \left(\frac{[\text{CLp}]}{K_8} \right)^2 + \left(\frac{[\text{P51p}]}{K_9} \right)^2 + \left(\frac{[\text{ELp}]}{K_{10}} \right)^2} \right) - k_4 [\text{ELm}] \quad (7)$$

$$\begin{aligned} \frac{d[\text{ELp}]}{dt} = p_4 [\text{ELm}] - \left(d_{e1} + \frac{d_{e2} [\text{COP1}] + d_{e3} [\text{COP1} : \text{PhyA}]}{[\text{COP1}] + [\text{COP1} : \text{PhyA}] + [\text{COP1} : \text{PhyB}] + [\text{COP1} : \text{Cry1}]} \right. \\ \left. + \frac{d_{e4} [\text{COP1} : \text{PhyB}] + d_{e5} [\text{COP1} : \text{Cry1}]}{[\text{COP1}] + [\text{COP1} : \text{PhyA}] + [\text{COP1} : \text{PhyB}] + [\text{COP1} : \text{Cry1}]} \right) [\text{ELp}] \end{aligned} \quad (8)$$

$$\begin{aligned} \frac{d[\text{PhyA}]}{dt} = (1 - \Theta_{\text{PhyA}}) A_{p3} - \frac{A_{m7} [\text{PhyA}]}{A_{k7} + [\text{PhyA}]} \\ - q_2 \Theta_{\text{PhyA}} [\text{PhyA}] - k_{mpac} \Theta_{\text{PhyA}} [\text{PhyA}] [\text{COP1}] \\ + k_d [\text{COP1} : \text{PhyA}] \end{aligned} \quad (9)$$

$$\frac{d[\text{PIFm}]}{dt} = \frac{v_5}{1 + \left(\frac{[\text{ELp}]}{K_{11}} \right)^2 + \left(\frac{[\text{Cry1}]}{K_{14}} \right)^2} - k_5 [\text{PIFm}] \quad (10)$$

$$\frac{d[\text{PIFp}]}{dt} = p_5 [\text{PIFm}] - d_{5D} (1 - \Theta_{\text{PhyA}}) [\text{PIFp}] - d_{5L} \Theta_{\text{PhyA}} [\text{PIFp}] \quad (11)$$

$$\frac{d[\text{HYPp}]}{dt} = g_1 + \frac{g_2[\text{PIFp}]^2}{K_{12}^2 + [\text{PIFp}]^2} \quad (12)$$

$$\begin{aligned} \frac{d[\text{PhyB}]}{dt} &= B_{p4} - \frac{B_{m8}[\text{PhyB}]}{B_{k8} + [\text{PhyB}]} \\ &\quad - k_{mpbc}\Theta_{\text{PhyB}}[\text{PhyB}][\text{COP1}] + k_d[\text{COP1} : \text{PhyB}] \end{aligned} \quad (13)$$

$$\begin{aligned} \frac{d[\text{Cry1}]}{dt} &= C_{p5} - \frac{C_{m9}[\text{Cry1}]}{C_{k9} + [\text{Cry1}]} \\ &\quad - k_{mcc}\Theta_{\text{Cry1}}[\text{Cry1}][\text{COP1}] + k_d[\text{COP1} : \text{Cry1}] \end{aligned} \quad (14)$$

$$\begin{aligned} \frac{d[\text{COP1}]}{dt} &= -k_{mpac}\Theta_{\text{PhyA}}[\text{PhyA}][\text{COP1}] + k_d[\text{COP1} : \text{PhyA}] \\ &\quad - k_{mpbc}\Theta_{\text{PhyB}}[\text{PhyB}][\text{COP1}] + k_d[\text{COP1} : \text{PhyB}] \\ &\quad - k_{mcc}\Theta_{\text{Cry1}}[\text{Cry1}][\text{COP1}] + k_d[\text{COP1} : \text{Cry1}] \\ &\quad + \frac{A_{m7}[\text{COP1} : \text{PhyA}]}{A_{k7} + [\text{COP1} : \text{PhyA}]} + q_2\Theta_{\text{PhyA}}[\text{COP1} : \text{PhyA}] \\ &\quad + \frac{B_{m8}[\text{COP1} : \text{PhyB}]}{B_{k8} + [\text{COP1} : \text{PhyB}]} + \frac{C_{m9}[\text{COP1} : \text{Cry1}]}{C_{k9} + [\text{COP1} : \text{Cry1}]} \end{aligned} \quad (15)$$

$$\begin{aligned} \frac{d[\text{COP1} : \text{PhyA}]}{dt} &= k_{mpac}\Theta_{\text{PhyA}}[\text{PhyA}][\text{COP1}] - k_d[\text{COP1} : \text{PhyA}] \\ &\quad - \frac{A_{m7}[\text{COP1} : \text{PhyA}]}{A_{k7} + [\text{COP1} : \text{PhyA}]} - q_2\Theta_{\text{PhyA}}[\text{COP1} : \text{PhyA}] \end{aligned} \quad (16)$$

$$\begin{aligned} \frac{d[\text{COP1} : \text{PhyB}]}{dt} &= k_{mpbc}\Theta_{\text{PhyB}}[\text{PhyB}][\text{COP1}] - k_d[\text{COP1} : \text{PhyB}] \\ &\quad - \frac{B_{m8}[\text{COP1} : \text{PhyB}]}{B_{k8} + [\text{COP1} : \text{PhyB}]} \end{aligned} \quad (17)$$

$$\begin{aligned} \frac{d[\text{COP1} : \text{Cry1}]}{dt} &= k_{mcc}\Theta_{\text{Cry1}}[\text{Cry1}][\text{COP1}] - k_d[\text{COP1} : \text{Cry1}] \\ &\quad - \frac{C_{m9}[\text{COP1} : \text{Cry1}]}{C_{k9} + [\text{COP1} : \text{Cry1}]} \end{aligned} \quad (18)$$

$$\begin{aligned} \frac{d[\text{GZp}]}{dt} &= G_{p6} - \left(d_{g1} + \frac{d_{g2}[\text{COP1}] + d_{g3}[\text{COP1} : \text{PhyA}]}{[\text{COP1}] + [\text{COP1} : \text{PhyA}] + [\text{COP1} : \text{PhyB}] + [\text{COP1} : \text{Cry1}]} \right. \\ &\quad \left. + \frac{d_{g4}[\text{COP1} : \text{PhyB}] + d_{g5}[\text{COP1} : \text{Cry1}]}{[\text{COP1}] + [\text{COP1} : \text{PhyA}] + [\text{COP1} : \text{PhyB}] + [\text{COP1} : \text{Cry1}]} \right) [\text{GZp}] \end{aligned} \quad (19)$$

Light activation term:

$$L_u = \left(q_{1u}([\text{PhyA}])\Theta_{\text{PhyA}} \right) + \left(q_{3u}([\text{PhyB}]) \log_{10}(\eta_1 I_{\text{red}} + 1) \Theta_{\text{PhyB}} \right) \\ + \left(q_{4u}([\text{Cry1}]) \log_{10}(\eta_2 I_{\text{blue}} + 1) \Theta_{\text{Cry1}} \right)$$

Here, $u = (\text{a or b})$ for CL and P97, respectively.

For Θ_{PhyA} , Θ_{PhyB} and Θ_{Cry1} values, the following conditions are followed-

$$\Theta_{\text{PhyA}} = \begin{cases} 1, & I_{\text{red}} \text{ or } I_{\text{blue}} \neq 0, \\ 0, & \text{otherwise} \end{cases}$$

$$\Theta_{\text{PhyB}} = \begin{cases} 1, & I_{\text{red}} \neq 0, \\ 0, & \text{otherwise} \end{cases}$$

$$\Theta_{\text{Cry1}} = \begin{cases} 1, & I_{\text{blue}} \neq 0, \\ 0, & \text{otherwise} \end{cases}$$